Competition with a Fixed Outside Option

Progress note — transcription of handwritten notes

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In this note we present a competition model where the customer chooses between two sellers and a known outside option of fixed quality p_0.

1 Elements of the model

- Two symmetric providers, indexed by $j \in \{A, B\}$.
- Two products, indexed by $i \in \{1, 2\}$.
- Success probabilities: $0 < p_1 < p_2 < 1$.
- Costs: $c_1 < c_2$.
- Revenue: $R > c_2$.
- Outside option quality $p_0 \in (0, 1)$, fixed and known to all agents.
- Discount factor $\gamma \in (0, 1)$ per calendar period.

2 Customer beliefs.

Recall that in the forgetting model, past observations are discounted by a factor ρ whenever a new outcome is observed. Thus, after k interactions with the seller, the total effective information satisfies

$$S_k + F_k = \rho(S_{k-1} + F_{k-1}) + 1,$$

with $S_0 + F_0 = 0$. Iterating this expression yields

$$S_k + F_k = 1 + \rho + \cdots + \rho^{k-1} = \frac{1 - \rho^k}{1 - \rho} \quad \to 1/(1 - \rho) \quad \text{as } k \to \infty.$$

To study the limiting behavior, we normalize the state as

$$x_s = (1 - \rho)S_k$$

$$x_f = (1 - \rho)F_k$$

Since $x_s + x_f = 1 - \rho^k \to 1$, we focus on the limiting boundary $x_s + x_f = 1$, which can be represented by a single variable $x = x_s$, with $x_f = 1 - x$. Then $x \in [0, 1]$ when $\rho \in [1/2, 1)$.

For each seller we will call this normalized state variable x_j for $j \in \{A, B\}$ that will represent the proportion of successes for provider j in the long run. The customer observes the state

$$x = (x_A, x_B),$$

Her beliefs are

$$\theta_A \mid x_A \sim \text{Beta}(1 + \frac{x_A}{1-\rho}, 1 + \frac{1-x_A}{1-\rho}), \tag{1}$$

$$\theta_B \mid x_B \sim \text{Beta}(1 + \frac{x_B}{1-\rho}, 1 + \frac{1-x_B}{1-\rho}). \tag{2}$$

Customer choice. The customer uses Thompson sampling (TS), drawing independently from the two belief distributions and choosing

$$y_t = \begin{cases} A, & \widetilde{\theta}_A > \max\{\widetilde{\theta}_B, p_0\}, \\ B, & \widetilde{\theta}_B > \max\{\widetilde{\theta}_A, p_0\}, \\ 0, & p_0 > \max\{\widetilde{\theta}_A, \widetilde{\theta}_B\}, \end{cases} \tag{3}$$

where 0 denotes the outside option. Ties have probability zero. The selection probabilities are

$$D_A(x_A, x_B; p_0) = \mathbb{P}(\widetilde{\theta}_A > \max\{\widetilde{\theta}_B, p_0\} \mid x_A, x_B), \tag{4}$$

$$D_B(x_A, x_B; p_0) = \mathbb{P}(\widetilde{\theta}_B > \max\{\widetilde{\theta}_A, p_0\} \mid x_A, x_B), \tag{5}$$

$$D_0(x_A, x_B; p_0) = \mathbb{P}(\widetilde{\theta}_A < p_0 \mid x_A)\mathbb{P}(\widetilde{\theta}_B < p_0 \mid x_B). \tag{6}$$

Hence $D_A + D_B + D_0 = 1$.

3 Timing, state transitions, and strategies

Timing.

1. The customer observes $x = (x_A, x_B)$, while each seller j observes only its own reputation x_j.
2. The customer chooses seller A, seller B, or the outside option using Thompson sampling.
3. If a seller is selected, it chooses a product $i \in \{1, 2\}$ according to its strategy $\sigma_j(\cdot \mid x_j)$ and receives profit $R - c_i$.
4. If a seller is selected, the customer observes the outcome, but not the product.
5. Only the selected seller's reputation is updated. If the outside option is chosen, neither reputation changes and both sellers receive zero profit.

Thus, forgetting occurs only when a new outcome is observed from a seller, whereas discounting applies in every calendar period.

State transitions. Superscripts s and f denote success and failure, respectively. The state transitions are

$$\begin{aligned} T_A^s(x) &= (\rho x_A + 1 - \rho, x_B), \\ T_A^f(x) &= (\rho x_A, x_B), \\ T_B^s(x) &= (x_A, \rho x_B + 1 - \rho), \\ T_B^f(x) &= (x_A, \rho x_B). \end{aligned} \tag{7}$$

For an individual seller, we write these updates as $T_\rho^s(x_j) = \rho x_j + 1 - \rho$ and $T_\rho^f(x_j) = \rho x_j$.

Strategies. We consider stationary mixed strategies that depend only on the seller's own reputation. For each seller $j \in \{A, B\}$, a strategy is a mapping

$$\sigma_j : [0,1] \to \Delta(\{1,2\}), \tag{8}$$

where $\sigma_j(i \mid x_j)$ denotes the probability that seller j chooses product i when selected at reputation x_j.

4 Oblivious equilibrium: Distribution

Oblivious approximation: Throughout this section, p_0 is fixed. It is a parameter of the model, rather than a conjecture determined by the equilibrium conditions. Suppose that each seller $j \in \{A, B\}$ assumes that its competitor's reputation follows a stationary distribution μ_{-j} on $[0,1]$, measured in calendar time before customer selection.

Instead of observing the competitor's current reputation, seller j computes its expected demand by averaging over this distribution. Write $D(x_j, y; p_0) = D_A(x_j, y; p_0)$, with the focal seller's reputation as the first argument. By symmetry, the same function applies to either seller. Then

$$d_j(x_j; \mu_{-j}, p_0) = \int_0^1 D(x_j, y; p_0)\, \mu_{-j}(dy). \tag{9}$$

Thus, each seller's demand depends only on its own reputation and the conjectured distribution of its competitor, given p_0. When optimizing, the seller treats this distribution as fixed.

Seller's problem: Let $U_j(x_j)$ denote seller j's optimal expected discounted profit before customer selection. We leave the dependence of U_j on the fixed parameter p_0 implicit. Given the conjectured distribution μ_{-j}, the Bellman equation is

$$\begin{aligned} U_j(x_j) = {} & \gamma[1 - d_j(x_j; \mu_{-j}, p_0)]U_j(x_j) \\ & + d_j(x_j; \mu_{-j}, p_0) \max_{i \in \{1,2\}} \Big\{ R - c_i \\ & \qquad + \gamma \left[p_i U_j(T_\rho^s(x_j)) + (1 - p_i) U_j(T_\rho^f(x_j)) \right] \Big\}. \end{aligned} \tag{10}$$

If the seller is not selected, either its competitor or the outside option is chosen. In both cases it receives no profit and its own reputation remains unchanged, which explains the term $\gamma(1 - d_j)U_j$. If selected, it chooses a product to maximize its current profit and expected future value.

Stationary distribution: Given a strategy σ_j, the probability of success conditional on seller j being selected is

$$p_{\sigma_j}(x_j) = \sigma_j(1 \mid x_j)p_1 + \sigma_j(2 \mid x_j)p_2. \tag{11}$$

Given a strategy profile $\sigma = (\sigma_A, \sigma_B)$, let $P^{\sigma, p_0}(x, C)$ denote the probability that the joint reputation state belongs to a measurable set $C \subseteq [0,1]^2$ in the next period, given the current state $x = (x_A, x_B)$. The transition probabilities are

$$\begin{aligned} P^{\sigma,p_0}(x, C) = {} & D_0(x_A, x_B; p_0)\mathbf{1}\{x \in C\} \\ & + \sum_{j \in \{A,B\}} D_j(x_A, x_B; p_0) \Big[p_{\sigma_j}(x_j)\mathbf{1}\{T_j^s(x) \in C\} \\ & \qquad + (1 - p_{\sigma_j}(x_j))\mathbf{1}\{T_j^f(x) \in C\} \Big]. \end{aligned} \tag{12}$$

The first term accounts for the outside option: both reputations remain unchanged. In particular, the probability that B sells is D_B, rather than $1 - D_A$.

Oblivious equilibrium: For the given p_0, a symmetric oblivious equilibrium is a mixed strategy σ^*, a reputation distribution μ^*, and a joint stationary distribution ν^* satisfying the following conditions. Here U_{μ^*} is the value from the seller's problem with $\mu_{-j} = \mu^*$ and the given p_0.

Optimality:

$$\begin{aligned}\operatorname{supp}\sigma^*(\cdot \mid x) \subseteq \arg\max_{i\in\{1,2\}} \Big\{ R - c_i + \gamma\big[p_i U_{\mu^*}(T_\rho^s(x)) \\ + (1-p_i)U_{\mu^*}(T_\rho^f(x))\big]\Big\}.\end{aligned} \tag{13}$$

Consistency:

$$\begin{aligned}\nu^* &= \nu^* P^{\sigma^*,p_0}, \\ \mu^*(C) = \nu^*(C\times[0,1]) &= \nu^*([0,1]\times C), \qquad \forall C \subseteq [0,1] \text{ measurable.}\end{aligned} \tag{14}$$

Here P^{σ^*,p_0} denotes the joint transition kernel of the actual two-seller game when both sellers follow σ^* and face the same fixed outside option p_0.

The first condition requires each seller to choose only optimal products given the conjectured distribution μ^*. The second requires this distribution to coincide with the marginal distributions generated when both sellers follow σ^* in the actual game, including periods in which the outside option is chosen. No independence between the two reputations is assumed.

5 Product choice as the number of sellers grows

Consider N sellers and one customer choice per calendar period, keeping p_0, $\rho < 1$, $\gamma < 1$, and all other parameters fixed. Write $D_j^{(N)}(x; p_0)$ for seller j's selection probability at the joint reputation state $x = (x_1, \ldots, x_N)$, and $U_N(x_j)$ for its oblivious value in the same Bellman equation as above.

Proposition. There exists a finite N_0 such that, for every $N \geq N_0$, product 1 is strictly optimal at every reputation, for any conjectured joint distribution of rival reputations. In particular, every symmetric distribution oblivious equilibrium satisfies $\sigma^*(1 \mid x_j) = 1$ for all $x_j \in [0,1]$ when $N \geq N_0$.

Proof. Consider that the customer selects seller j when its TS draw exceeds both p_0 and every rival's draw. Ignoring p_0 can only increase this probability, so

$$\begin{aligned}D_j^{(N)}(x; p_0) &= \mathbb{P}\left(\widetilde{\theta}_j > \max\{p_0, \max_{k\neq j}\widetilde{\theta}_k\} \mid x\right) \\ &\leq \mathbb{P}(\widetilde{\theta}_j > \widetilde{\theta}_k \text{ for all } k \neq j \mid x).\end{aligned} \tag{15}$$

Now, write $\widetilde{\theta}_j(z)$ for seller j's TS draw at reputation z. If $x' > x$, then

$$\mathbb{P}(\widetilde{\theta}_j(x') > t) \geq \mathbb{P}(\widetilde{\theta}_j(x) > t), \qquad t \in (0,1). \tag{*}$$

We prove (*) below. The most favorable case for seller j is therefore to have reputation 1, while every rival has reputation 0. Hence

$$\mathbb{P}(\widetilde{\theta}_j > \widetilde{\theta}_k \text{ for all } k \neq j \mid x) \leq \mathbb{P}(\widetilde{\theta}_j(1) > \widetilde{\theta}_k(0) \text{ for all } k \neq j). \tag{16}$$

Consider that $\widetilde{\theta}_j(1) \sim \text{Beta}(a, 1)$ and $\widetilde{\theta}_k(0) \sim \text{Beta}(1, a)$, where $a = 1 + 1/(1-\rho)$. For $t \in (0,1)$,

$$\mathbb{P}(\widetilde{\theta}_k(0) < t) = \int_0^t a(1-s)^{a-1}\, ds = 1 - (1-t)^a. \tag{17}$$

Since the TS draws are independent conditional on reputations, averaging over seller j's draw gives

$$\begin{aligned} &\mathbb{P}(\widetilde{\theta}_j(1) > \widetilde{\theta}_k(0) \text{ for all } k \neq j) \\ &\quad = \int_0^1 \big[\mathbb{P}(\widetilde{\theta}_k(0) < t)\big]^{N-1} a t^{a-1}\, dt \\ &\quad = \int_0^1 \big[1 - (1-t)^a\big]^{N-1} a t^{a-1}\, dt \\ &\quad = \mathbb{E}\Big[\big(1 - (1 - \widetilde{\theta}_j(1))^a\big)^{N-1}\Big]. \end{aligned} \tag{18}$$

The bound is valid at every joint reputation state, so

$$\sup_{x \in [0,1]^N} D_j^{(N)}(x; p_0) \leq \mathbb{E}\Big[\big(1 - (1 - \widetilde{\theta}_j(1))^a\big)^{N-1}\Big]. \tag{19}$$

Since $\widetilde{\theta}_j(1) \in (0,1)$ almost surely,

$$\big(1 - (1 - \widetilde{\theta}_j(1))^a\big)^{N-1} \longrightarrow 0 \quad \text{almost surely,} \tag{20}$$

and this expression lies between 0 and 1. By dominated convergence,

$$\begin{aligned} &\lim_{N\to\infty} \mathbb{E}\Big[\big(1 - (1 - \widetilde{\theta}_j(1))^a\big)^{N-1}\Big] \\ &\quad = \mathbb{E}\left[\lim_{N\to\infty} \big(1 - (1 - \widetilde{\theta}_j(1))^a\big)^{N-1}\right] = 0. \end{aligned} \tag{21}$$

It follows that

$$\sup_{x \in [0,1]^N} D_j^{(N)}(x; p_0) \longrightarrow 0. \tag{22}$$

The seller's oblivious demand averages $D_j^{(N)}$ over the conjectured joint distribution of rival reputations, so the same bound applies to any such distribution, including correlated reputations.

Now recall that profit per sale is bounded by $R - c_1$. Therefore,

$$\begin{aligned} 0 \leq U_N(x_j) &\leq \sum_{t=0}^{\infty} \gamma^t (R - c_1) \sup_{x \in [0,1]^N} D_j^{(N)}(x; p_0) \\ &= \frac{R - c_1}{1 - \gamma} \sup_{x \in [0,1]^N} D_j^{(N)}(x; p_0) \longrightarrow 0. \end{aligned} \tag{23}$$

Using this in the optimality condition, and writing $\Delta c = c_2 - c_1 > 0$ and $\Delta p = p_2 - p_1 > 0$, we obtain

$$\begin{aligned} &\lim_{N\to\infty} \Big\{ -\Delta c + \gamma \Delta p \big[U_N(T_\rho^s(x_j)) - U_N(T_\rho^f(x_j))\big] \Big\} \\ &\quad = -\Delta c + \gamma \Delta p (0 - 0) = -\Delta c < 0. \end{aligned} \tag{24}$$

The convergence is uniform in reputation and rival conjectures. Thus, product 1 is strictly optimal at every reputation for all sufficiently large N. □

Proof of (*)**.** Let f_x denote the posterior density at reputation x. For $x' > x$,

$$\frac{f_{x'}(t)}{f_x(t)} = C\left(\frac{t}{1-t}\right)^{(x'-x)/(1-\rho)}, \qquad C > 0, \tag{25}$$

where C does not depend on t. This ratio increases continuously from 0 to infinity. Hence the densities cross once, at some $t_* \in (0,1)$: $f_{x'} \le f_x$ below t_* and $f_{x'} \ge f_x$ above it. Since both densities integrate to 1,

$$\begin{aligned} &\int_0^t \big[f_{x'}(s) - f_x(s)\big]\, ds \le 0, \qquad t \le t_*, \\ &\int_0^t \big[f_{x'}(s) - f_x(s)\big]\, ds = -\int_t^1 \big[f_{x'}(s) - f_x(s)\big]\, ds \le 0, \quad t \ge t_*. \end{aligned} \tag{26}$$

Taking complements gives

$$\mathbb{P}(\widetilde{\theta}_j(x') > t) = 1 - \int_0^t f_{x'}(s)\, ds \ge 1 - \int_0^t f_x(s)\, ds = \mathbb{P}(\widetilde{\theta}_j(x) > t), \tag{27}$$

which proves (*). □